Programming Fundamentals

WEEK #4

Arab Academy
for Science, Technology and Maritime Transport

Lecture Agenda

- **Arithmetic Operators**
- **Operators Precedence**
- **Converting Algebraic Expressions to Programming Statements**
- **Combined Assignment Operators**
- **Mathematical Library Functions**

Arithmetic Operators

There are many operators for manipulating numeric values and performing arithmetic operations

Table 2-9 Fundamental Arithmetic Operators

Operator	Meaning	Type	Example
+	Addition	Binary	`total = cost + tax;`
–	Subtraction	Binary	`cost = total - tax;`
*	Multiplication	Binary	`tax = cost * rate;`
/	Division	Binary	`salePrice = original / 2;`
%	Modulus	Binary	`remainder = value % 3;`

Arithmetic Operators

WARNING! When both operands of a division statement are integers, the statement will perform *integer division*. This means the result of the division will be an integer as well. If there is a remainder, it will be discarded. For example, in the following statement, `parts` is assigned the value 5:

```
parts = 17 / 3;
```

This may seem like an annoyance, but it can actually be useful in some programs. Remember, C++ gives you the tools to solve just about any problem! If you want to make sure a statement, like the one shown above, performs regular division, express one of the numbers as a floating point. Here is an example:

```
parts = 17.0 / 3;
```

In the statement above, since 17.0 is interpreted as a floating point number, the division operation will return a floating point number. The result of the division is 5.66667.

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EXAMPLES

11 / 4 = 2

11 % 4 = 3

11.0 / 4.0 = 2.75

11 / 4.0 = 2.75

15 / 3 = 5

3 / 6 = 0

6 % 3 = 0

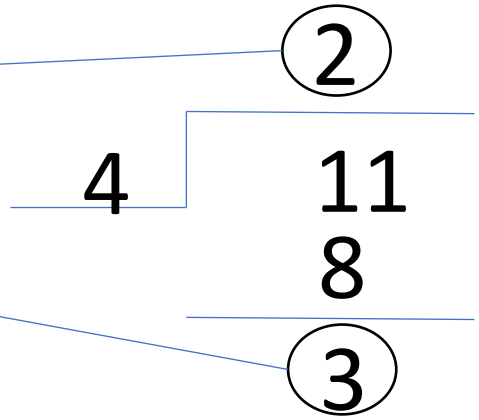
5 % 5 = 0

3 % 5 = 3

4.0 % 5 = error

5 % 4.0 = error

6.0 % 5.0 = error



Operators Precedence

Table 3-3 Associativity of Arithmetic Operators

Operator	Associativity
* / %	Left to right
+ -	Left to right

Table 3-4 More Simple Expressions and Their Values

Expression	Value
`(5 + 2) * 4`	28
`10 / (5 - 3)`	5
`8 + 12 * (6 - 2)`	56
`(4 + 17) % 2 - 1`	0
`(6 - 3) * (2 + 7) / 3`	9

Converting Algebraic Expressions to Programming Statements

Table 3-6 Algebraic and C++ Expressions

Algebraic Expression	C++ Expression
$y = 3\frac{x}{2}$	y = x / 2 * 3;
$z = 3bc + 4$	z = 3 * b * c + 4;
$a = \frac{3x + 2}{4a - 1}$	a = (3 * x + 2) / (4 * a - 1)

Combined Assignment Operators

For convenience, C++ offers a special set of operators designed specifically combined assignment operators as follow.

Table 3-9

Operator	Example Usage	Equivalent to
`+=`	`x += 5;`	`x = x + 5;`
`-=`	`y -= 2;`	`y = y - 2;`
`*=`	`z *= 10;`	`z = z * 10;`
`/=`	`a /= b;`	`a = a / b;`
`%=`	`c %= 3;`	`c = c % 3;`

Example:
Rewrite the following statements with combined assignment operators

Statements with Assignment Operator	Statements with Combined Assignment Operator
`x = x + 5;`	
`total = total + subtotal;`	
`dist = dist / rep;`	
`ppl = ppl * period;`	

EXAMPLE

```
int X=5 , Y=7, Z=10;
Z+=X % Y + Y / X;
```

Solution	Another Solution
int X=5 , Y=7, Z=10;	int X=5 , Y=7, Z=10;
Z+=X % Y + Y / X;	Z+=X % Y + Y / X;
Z = Z + (X % Y + Y / X);	Z+= 5 % 7 + 7 / 5;
Z = 10 + (5 % 7 + 7 / 5)	Z+= 5 + 1
Z = 10 + (5 + 1) = 16	Z = 10 + 6 = 16

Mathematical Library Functions

The C++ library provides several functions for performing complex mathematical operations.

Table 3-13

Function	Example	Description
`abs`	`y = abs(x);`	Returns the absolute value of the argument. The argument and the return value are integers.
`cos`	`y = cos(x);`	Returns the cosine of the argument. The argument should be an angle expressed in radians. The return type and the argument are `doubles`.
`exp`	`y = exp(x);`	Computes the exponential function of the argument, which is `x`. The return type and the argument are `doubles`.
`fmod`	`y = fmod(x, z);`	Returns, as a `double`, the remainder of the first argument divided by the second argument. Works like the modulus operator, but the arguments are `doubles`. (The modulus operator only works with integers.) Take care not to pass zero as the second argument. Doing so would cause division by zero.
`log`	`y = log(x);`	Returns the natural logarithm of the argument. The return type and the argument are `doubles`.
`log10`	`y = log10(x);`	Returns the base-10 logarithm of the argument. The return type and the argument are `doubles`.
`sin`	`y = sin(x);`	Returns the sine of the argument. The argument should be an angle expressed in radians. The return type and the argument are `doubles`.
`sqrt`	`y = sqrt(x);`	Returns the square root of the argument. The return type and argument are `doubles`.
`tan`	`y = tan(x);`	Returns the tangent of the argument. The argument should be an angle expressed in radians. The return type and the argument are `doubles`.

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EXAMPLES

R1= $\dfrac{-b-\sqrt{b^2-4ac}}{2a}$ R1=(-b-sqrt(b*b-4*a*c))/(2*a)

$Z = |a-b| \div c^4$ Z=fabs(a-b)/pow(c,4)

$$z = \frac{e^{x} - e^{-x}}{e^{2x}}$$

Z = (exp(x) – exp(-x))/exp(2*x)

EXAMPLES

- sqrt(9)=3.0
- pow(2,3.0)=8.0
- fabs(-9.0) =9.0
- ceil(3.2)=4.0
- floor(3.7) = 3.0

Exercise

Write C++ expressions for the following algebraic expressions:

$$a = 12x$$

$$y = x^4$$

$$g = \frac{h + 12}{4k}$$

Mathematical Library Functions

Program 3-25

$$c = \sqrt{a^2 + b^2}$$

```
// This program asks for the lengths of the two sides of a
// right triangle. The length of the hypotenuse is then
// calculated and displayed.
#include <iostream>
#include <cmath>     // For the sqrt and pow functions
using namespace std;

int main()
{
   double a, b, c;

   cout << "Enter the length of side a: ";
   cin >> a;
   cout << "Enter the length of side b: ";
   cin >> b;
   c = sqrt(pow(a, 2.0) + pow(b, 2.0));
   cout << "The length of the hypotenuse is ";
   cout << c << endl;
   return 0;
}
```

Program Output with Example Input Shown in Bold

```
Enter the length of side a: 5.0 [Enter]
Enter the length of side b: 12.0 [Enter]
The length of the hypotenuse is 13
```

Programming Challenges

a) Write a program ask the user to enter an angle, then calculates and displays the sin, cos and tan of this angle.

b) Write a program that calculates the roots of the quadratic equation. $ax^2 + bx + c = 0$.
The user must enter the values of a, b and c then the program displays the two roots using the following equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
        float  a, b, c, root_1, root_2;
        cout<<“enter the equation coefficients:\n”;
        cin>>a>>b>>c;
        root_1 = (-b - sqrt(pow(b,2) – 4 * a * c)) / (2 * a);
        root_2 = (-b +sqrt(pow(b,2) – 4 * a * c)) / (2 * a);
        cout<<“root 1 =”<<root_1<<endl;
        cout<<“root 2 =”<<root_2;
        return 0;
}
```

Thank You